QUASI-COHERENCE OF THE DERIVED PULLBACK AND ITS COMPATIBILITY WITH THE MODULE FUNCTOR ON AFFINES

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Allowed standard prerequisites (editorial).

- On affines the derived pullback agrees with derived tensor with the ring map on module complexes, up to the K-flat replacement construction. invoked at proof lines 607-609. Gives the commutativity of the square of functors.
- On an affine scheme the derived category of quasi-coherent modules is equivalent to the derived category of modules. invoked at proof lines 613-615. Translates E into a complex of A-modules and identifies the pullback.

Source reference. Original lemma 08DV: sources/stacks-perfect/pinned-original-source.tex, source lines 544–544. Cited at proof lines 607-608 for the commutativity of the first diagram; the localization and conclusion are inside the counted span.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Cited at proof lines 613-615 to represent E by a complex of A-modules; the conclusion of part (1) is inside the counted span.

Lemma 0.2. *Let $f : Y \to X$ be a morphism of schemes.*

(1) *The functor Lf^* sends $D_{QCoh}(\mathcal{O}_X)$ into $D_{QCoh}(\mathcal{O}_Y)$.*
(2) *If X and Y are affine and f is given by the ring map $A \to B$, then the diagram*

$$\begin{array}{ccc} D(B) & \longrightarrow & D_{QCoh}(\mathcal{O}_Y) \\ {\scriptstyle -\otimes^{\mathbf{L}}_A B}\uparrow & & \uparrow{\scriptstyle Lf^*} \\ D(A) & \longrightarrow & D_{QCoh}(\mathcal{O}_X) \end{array}$$

commutes.

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Proof. We first prove the diagram

$$\begin{array}{ccc} D(B) & \longrightarrow & D(\mathcal{O}_Y) \\ {\scriptstyle -\otimes^{\mathbf{L}}_A B}\uparrow & & \uparrow{\scriptstyle Lf^*} \\ D(A) & \longrightarrow & D(\mathcal{O}_X) \end{array}$$

commutes. This is clear from Lemma 08DV and the constructions of the functors in question. To see (1) let E be an object of $D_{QCoh}(\mathcal{O}_X)$. To see that Lf^*E has quasi-coherent cohomology sheaves we may work locally on X. Note that Lf^* is compatible with restricting to open subschemes. Hence we can assume that f is a morphism of affine schemes as in (2). Then we can apply Lemma 06Z0 to see that E comes from a complex of A-modules. By the commutativity of the first diagram of the proof the same holds for Lf^*E and we conclude (1) is true. □